

Host and Pathogen Glycosaminoglycan-Binding Proteins Modulate Antimicrobial Peptide Responses in *Drosophila melanogaster*[†]

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During group B streptococcal infection, the alpha C protein (ACP) on the bacterial surface binds to host cell surface heparan sulfate proteoglycans (HSPGs) and facilitates entry of bacteria into human epithelial cells. Previous studies in a *Drosophila melanogaster* model showed that binding of ACP to the sulfated polysaccharide chains (glycosaminoglycans) of HSPGs promotes host death and is associated with higher bacterial burdens. We hypothesized that ACP-glycosaminoglycan binding might determine infection outcome by altering host responses to infection, such as expression of antimicrobial peptides. As glycosaminoglycans/HSPGs also interact with a number of endogenous secreted signaling molecules in *Drosophila*, we examined the effects of host and pathogen glycosaminoglycan/HSPG-binding structures in host survival of infection and antimicrobial peptide expression. Strikingly, host survival after infection with wild-type streptococci was enhanced among flies overexpressing the endogenous glycosaminoglycan/HSPG-binding morphogen Decapentaplegic—a transforming growth factor β -like *Drosophila* homolog of mammalian bone morphogenetic proteins—but not by flies overexpressing a mutant, non-glycosaminoglycan-binding Decapentaplegic, or the other endogenous glycosaminoglycan/HSPG-binding morphogens, Hedgehog and Wingless. While ACP-glycosaminoglycan binding was associated with enhanced transcription of peptidoglycan recognition proteins and antimicrobial peptides, Decapentaplegic overexpression suppressed transcription of these genes during streptococcal infection. Further, the glycosaminoglycan-binding domain of ACP competed with Decapentaplegic for binding to the soluble glycosaminoglycan heparin in an *in vitro* assay. These data suggest that, in addition to promoting bacterial entry into host cells, ACP competes with Decapentaplegic for binding to glycosaminoglycans/HSPGs during infection and that these bacterial and endogenous glycosaminoglycan-binding structures determine host survival and regulate antimicrobial peptide transcription.

Heparan sulfate (HS) proteoglycans (HSPGs) are a superfamily of cell surface and extracellular matrix macromolecules that consist of core proteins linked to heparan sulfate glycosaminoglycan (GAG) chains. The GAG chains are synthesized by polymerases and subsequently modified by various sulfotransferases (16). Biochemical and genetic studies *in vivo* have demonstrated that HSPGs play crucial roles in cell-to-cell signaling and morphogenesis by controlling the distribution of secreted morphogens, including members of the Wnt/Wingless (Wnt/Wg), Hedgehog (Hh), transforming growth factor β (TGF- β), and fibroblast growth factor (FGF) families (18, 23). HSPGs have also been shown to be engaged by a diverse group of microorganisms that may usurp GAGs to facilitate their own cellular invasion during pathogenesis (7, 31).

Streptococcus agalactiae (group B *Streptococcus* [GBS]) is the most common cause of bacterial sepsis and meningitis in newborns (21, 32, 33). Alpha C protein (ACP), an important vir-

ulence determinant on the GBS surface, contains a GAG/HSPG-binding site at the junction of its N-terminal domain and the first of a series of tandem repeats (3, 30). Deficiency in ACP attenuates GBS virulence in neonatal mice (27). ACP facilitates GBS entry into and transcytosis across human epithelial cells (4, 5, 10), and ACP-GAG binding promotes GBS virulence in a *Drosophila melanogaster* infection model (6). However, little is known about the effects of ACP-GAG binding on specific host immune responses. We hypothesized that bacterial ACP and endogenous GAG-binding structures might alter innate host responses to infection, such as induction of antimicrobial peptides (AMPs).

We used a genetically amenable *Drosophila* model to test this hypothesis. The model has two advantages. First, *Drosophila* flies have a relatively simple immune system. Host responses to pathogens in *Drosophila* rely mainly on several components of the innate immune system that have mammalian parallels involving products of homologous genes. These components include phagocytosis by hemocytes and AMPs secreted by fat body cells (13, 25). *Drosophila* AMP production is regulated by two distinct signaling pathways: the Toll pathway, largely activated by fungi and Gram-positive bacteria, and the immune deficiency (Imd) pathway, mainly activated by Gram-negative bacteria (25, 26). Second, *Drosophila* flies use a relatively small number of genes that are highly homologous to the mammalian GAG/HSPG synthesis machinery to assemble a repertoire of GAG/HSPG structures that is similar to that of

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mammals (16). Specifically, the *Drosophila* genome contains three genes encoding HSPG core protein homologs (two glypicans, *dally* and *dally-like protein* [*dlp*], and one syndecan, *syndecan* [*sd*]), three genes encoding GAG polymerases (mammalian EXT1 homolog, *tout-velu* [*ttv*]; EXT2 homolog, *sister of ttv* [*sotv*]; and EXTL3 homolog, *brother of ttv* [*botv*]), and one gene encoding an *N*-deacetylase/*N*-sulfotransferase (NDST; *sulfateless* [*sfl*]). The biosynthesis of heparan sulfate GAG chains on the core proteins is catalyzed by Ttv and Sotv as copolymerases after chain initiation by Botv. The GAG chains then undergo a series of modifications, such as *N*-deacetylation and *N*-sulfation by Sfl (29).

The aim of these studies was to determine whether GAG-binding structures influence AMP responses in the *Drosophila* infection model. We found that, compared with wild-type GBS, a mutant GBS strain expressing a variant of ACP with a single amino acid change (R185A) that specifically reduces ACP-GAG binding had attenuated virulence and induced less transcription of AMPs and peptidoglycan recognition proteins (PGRPs) SA and SD—key microbe-recognizing receptors that may function in the Toll pathway—in both wild-type and GAG/HSPG-deficient flies. Competition enzyme-linked immunosorbent assay (ELISA) showed that ACP with intact GAG-binding ability could compete with Decapentaplegic (Dpp), a TGF- β -like *Drosophila* homolog of mammalian bone morphogenetic proteins (35), for binding to the GAG heparin *in vitro*, whereas a mutant version of ACP with reduced GAG-binding affinity could not. Overexpression of GAG-binding Dpp in hemocytes protected flies from lethal GBS infection and suppressed PGRP and AMP induction. These data suggest that (i) in addition to promoting bacterial entry into host cells, ACP competes with Dpp for binding to GAGs/HSPGs during infection and (ii) these GAG-binding structures determine host survival and regulate AMP transcription.

MATERIALS AND METHODS

Bacterial infection. A909 is a type Ia GBS wild-type strain that expresses ACP, and R185A is an A909 strain with a point mutation that expresses an ACP variant with diminished GAG-binding affinity but preserved growth rate, capsule production, and binding to ACP-specific antibodies. These characteristics suggest that behavioral differences between the mutant and wild-type strains are specific to the point mutation R185A (5). For infection experiments, bacteria were grown to an optical density of 0.3 at 650 nm and then concentrated 10-fold to $\sim 2 \times 10^9$ CFU/ml; the exact bacterial concentration was determined in each experiment. Adult male flies (2 to 5 days old) were anesthetized with CO₂ and then pricked in the dorsal thorax underneath the wing with a fine needle that had been dipped in Todd-Hewitt broth (THB) or a concentrated solution of GBS in THB. After infection, flies were incubated at 29°C in vials with food, and fly survival was monitored over the following 4 days. Cumulative survival curves were derived, and the median survival time for each group was determined by Kaplan-Meier survival analysis. Survival curves were compared by use of a log-rank test. A typical experiment included at least 30 flies per treatment group (6).

To determine the bacterial burden in whole flies, 10 infected flies per group were ground in 200 μ l of phosphate-buffered saline (PBS) with 0.025% Triton X-100 and then added to 800 μ l of PBS with 0.025% Triton X-100. Dilutions in PBS were plated onto THB agar and incubated overnight at 37°C for CFU determination.

Fly strains and genetics. All uninfected flies were maintained at 25°C and raised on standard *Drosophila* medium unless specified otherwise. The HSPG mutant alleles of *dally*, *dlp*, and *sd* and the GAG polymerase mutant alleles of *ttv* and *sotv* used in these studies have been described previously (6). They are as follows: (i) *dally* plus *dally-like protein*, *dally80 dlyA187*, with null alleles for both of these HSPGs; *dally80* has a *dally* deletion from amino acid positions –724 to

415, while *dlyA187* has a deletion of 26 nucleotides resulting in a reading-frame shift from amino acid 205 and therefore lacks part of the cysteine-rich region, the entire GAG attachment domain, and the glycosylphosphatidylinositol (GPI)-anchoring signal (20); (ii) *sd*, Bloomington stock center number 8585=w[*]; Df(2R)48, P{w[+mC]=Ubi-Sara.J}2/CyO, P{w[+mC]=ActGFP}JMR1, with an *sd* gene deficiency (22); and (iii) *ttv* and *sotv*, *FRTG13 ttv00681 sotv 1.8.1/CyO* (kind gift of Rahul Warrior). The *sotv 1.8.1* allele generates an Sotv protein truncated at residue 184 and lacking the catalytic motif (11).

The *sulfateless* mutant allele *sfl^{9B4}*, Hml Δ -Gal4, UAS-Hh, and UAS-Wg flies were kindly provided by Norbert Perrimon. UAS-Dpp-GFP flies were kindly provided by Kristi Wharton. UAS-HA-Dpp and UAS-HA-Dpp^{AN} flies were kindly provided by Hiroshi Nakato. Unless specified otherwise, *yw* flies were used as controls in all assays. Heterozygous mutants were generated by crossing mutant flies with *yw* flies.

Quantitative PCR. Approximately 15 flies were homogenized in TRIzol reagent (Invitrogen) to isolate total RNA, which was then purified with an RNeasy minikit (Qiagen) with on-column DNase digestion. cDNA was synthesized from 1 μ g of the purified RNA with a QuantiTect (Qiagen) reverse transcription kit. Quantitative PCR (qPCR) was performed in a standard manner according to the manufacturer's instructions. Diluted cDNA mixtures prepared as described above were used as templates (1 μ l each) and run on an Applied Biosystems 7300 real-time PCR system. For each condition, cDNA was obtained from three independent RNA preparations for repeating and averaging. The relative differences of mRNA expression levels in four AMP genes (*cecropin*, *defensin*, *dipthericin*, and *drosomycin*) and other genes were quantified by comparison of their levels with standard curves constructed with the corresponding series dilution of cDNA mixtures and were normalized to the expression level of the reference gene *rp49*. mRNA levels measured in wild-type flies 24 h after infection with A909 were set equal to 1, and the values obtained from other samples were expressed relative to these control values. Student's *t* test was used for statistical analysis of relative mRNA levels. The primers used for qPCR are shown in Table S1 in the supplemental material.

Surface plasmon resonance (SPR) assay. The sensor chip was incubated with HS-polyethylene glycol (PEG)-COOH (2 mg/ml) for 24 to 48 h and then washed with distilled H₂O. The carboxyl groups of the PEG were activated by *N*-hydroxysuccinimide (NHS)–1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDC) and neutravidin (1 mg/ml) was immobilized. Biotinylated HS was prepared and injected on the flow cell with a response unit change of ~ 200 . A chip without biotin-HS was used to control for nonspecific binding to the chip. The chip was then tested with 400 nM FGF-1, a known HS-binding protein. The interaction studies were monitored at a flow rate of 40 μ l/min at 25°C. The kinject method was used to minimize dispersion, with a 3-min association time and variable dissociation. After each injection, the sensor surface was regenerated with 50 mM NaOH–1 M NaCl. The control flow cell was used for subtraction from the experimental flow cells.

Preparation of heparin-coated plates. Heparin-coated plates were prepared by adding 50 μ l of heparin (2 mg/ml in 0.1 M carbonate buffer; pH 9.6; Sigma) to 384-well plates (untreated polystyrene; Corning) and incubating overnight at 4°C. Plates were subsequently washed five times with 100 μ l of PBS–0.05% Tween 20 (PBS-T). Free sites of the plate were blocked for 2 h at room temperature with 50 μ l of blocking buffer (incubation buffer): PBS-T containing 1% bovine serum albumin, 0.05% casein (from stock of 2% casein in 0.1 M NaOH), 5 mM thimerosal, and 2% sucrose. The plate was then washed five times with 100 μ l of PBS-T.

Competition ELISA. Recombinant *Drosophila* Dpp protein (R&D Systems) was diluted to a concentration of 5×10^{-3} μ g/ μ l in incubation buffer and added in a 20- μ l volume to each well of a heparin-coated 384-well plate. Competitor proteins included full-length ACP, D2-R (a truncated ACP construct containing the GAG-binding region of ACP), and a mutant D2-R protein containing the R185A mutation that confers decreased GAG-binding affinity. These proteins were expressed as described elsewhere (3, 5, 10). Serial dilutions of competitor proteins ranging from 9×10^{-3} M to 9×10^{-6} M were made in incubation buffer. Diluted protein competitors were added to the appropriate wells in 20- μ l volumes to reach a 40- μ l total volume per well. Plates were then incubated overnight at 4°C. The wells were washed five times with PBS-T, and the plates were incubated overnight at 4°C with 40 μ l of monoclonal antibody to *Drosophila* Dpp (R&D Systems) diluted 1:750 in incubation buffer. After five more washes with PBS-T, alkaline phosphatase-conjugated goat anti-mouse IgG (MP Bio-medicals, LLC) diluted 1:2,000 in incubation buffer was added, and plates were incubated for 2 h at room temperature. The absorbance at 405 nm was measured after incubation for 15 min with phosphatase substrate (Sigma).

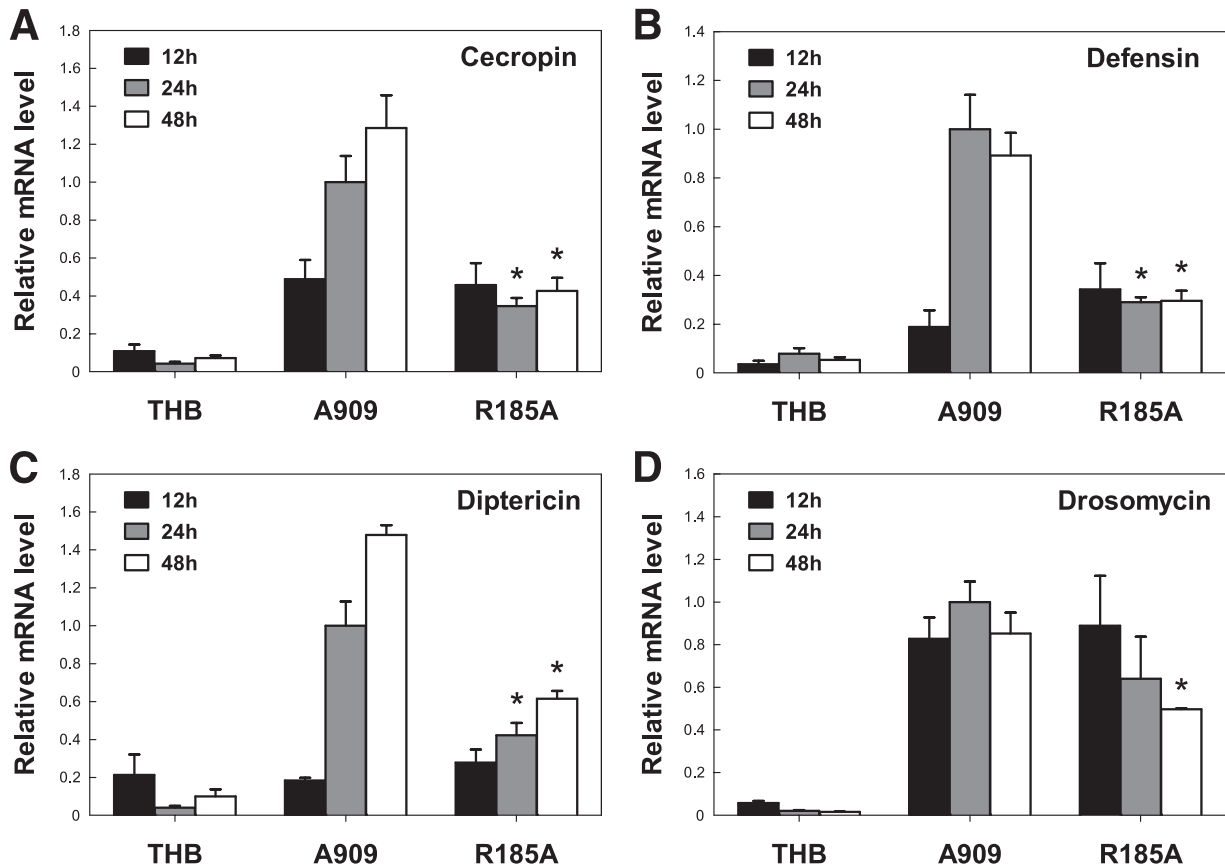


FIG. 1. Induction of AMP gene transcription after GBS infection over time. (A to D) Wild-type *Oregon R* flies were treated by broth injury (THB), infection with wild-type GBS strain A909, or infection with mutant GBS strain R185A and then harvested 12 h (black bars), 24 h (gray bars), and 48 h (white bars) postinfection. Induction of AMP gene transcription of *cecropin* (A), *defensin* (B), *dipteracin* (C), and *drosomycin* (D) in the flies after infection was detected by quantitative PCR. Relative mRNA levels of each gene were calculated according to the corresponding standard curve and normalized to the constitutively expressed ribosomal gene *rp49*. Each bar represents the mean and standard error of the mean of relative mRNA levels detected in triplicate independent fly samples. Comparisons of each AMP level at different time points after A909 and R185A infection were analyzed by Student's *t* test. *, $P < 0.05$.

RESULTS

GAG-binding site of ACP induces AMP transcription in *Drosophila*. We first examined AMP responses to ACP-GAG binding, using type Ia GBS wild-type strain A909 and a mutant strain, R185A, derived from A909. This mutant strain expresses a variant of ACP with a single amino acid change (R185A) that specifically reduces both ACP-GAG binding and GBS entry into human cells (5). We chose *cecropin*, *defensin*, *dipteracin*, and *drosomycin* as representative AMP genes and measured their transcription in wild-type *Oregon R* flies at 12 h, 24 h, and 48 h after infection. The relative mRNA levels of most of these AMPs were highly induced from 12 h to 24 h after infection with A909 and persisted or continued to rise until at least 48 h after infection. Specifically, compared with the levels of induction for sterile THB-treated flies, A909-infected flies showed inductions at 12 h and 24 h of 4.5-fold and 23.4-fold, respectively, for *cecropin*; 5.3-fold and 12.6-fold, respectively, for *defensin*; 0.9-fold and 25.0-fold, respectively, for *dipteracin*; and 14.5-fold and 49.1-fold, respectively, for *drosomycin*. At 48 h, inductions were 17.8-fold for *cecropin*, 16.5-fold for *defensin*, 14.8-fold for *dipteracin*, and 54.4-fold for *drosomycin*. Infection with strain R185A induced lower levels of AMP

transcription (Fig. 1). Specifically, compared with the levels of induction for sterile THB-treated flies, R185A-treated flies showed inductions at 12 h, 24 h, and 48 h that were 4.2-fold, 8.1-fold, and 5.9-fold, respectively, for *cecropin*; 9.6-fold, 3.7-fold, and 5.5-fold, respectively, for *defensin*; 1.3-fold, 10.6-fold, and 6.2-fold, respectively, for *dipteracin*; and 15.5-fold, 31.4-fold, and 31.7-fold, respectively, for *drosomycin*. AMP genes in the control flies (pricked with a needle dipped in sterile THB) were transcribed consistently at very low levels over time (Fig. 1). These data suggest that mutation of the GAG-binding site in ACP reduced the mutant strain's ability to induce AMP genes after infection.

PGRP-SA and PGRP-SD, peptidoglycan recognition components that may function in the Toll pathway, are dramatically induced by GBS challenge. The Toll pathway largely responds to Gram-positive bacterial infection and fungal infection, activating the transcription of *drosomycin* and contributing to the induction of *defensin* (36). We examined the transcription of individual Toll pathway members 24 h after infection with the Gram-positive bacterial pathogen GBS (Fig. 2A). We also looked at transcription of PGRP-SA and PGRP-SD, which are critically involved in sensing Gram-positive bacterial infection and may activate Toll

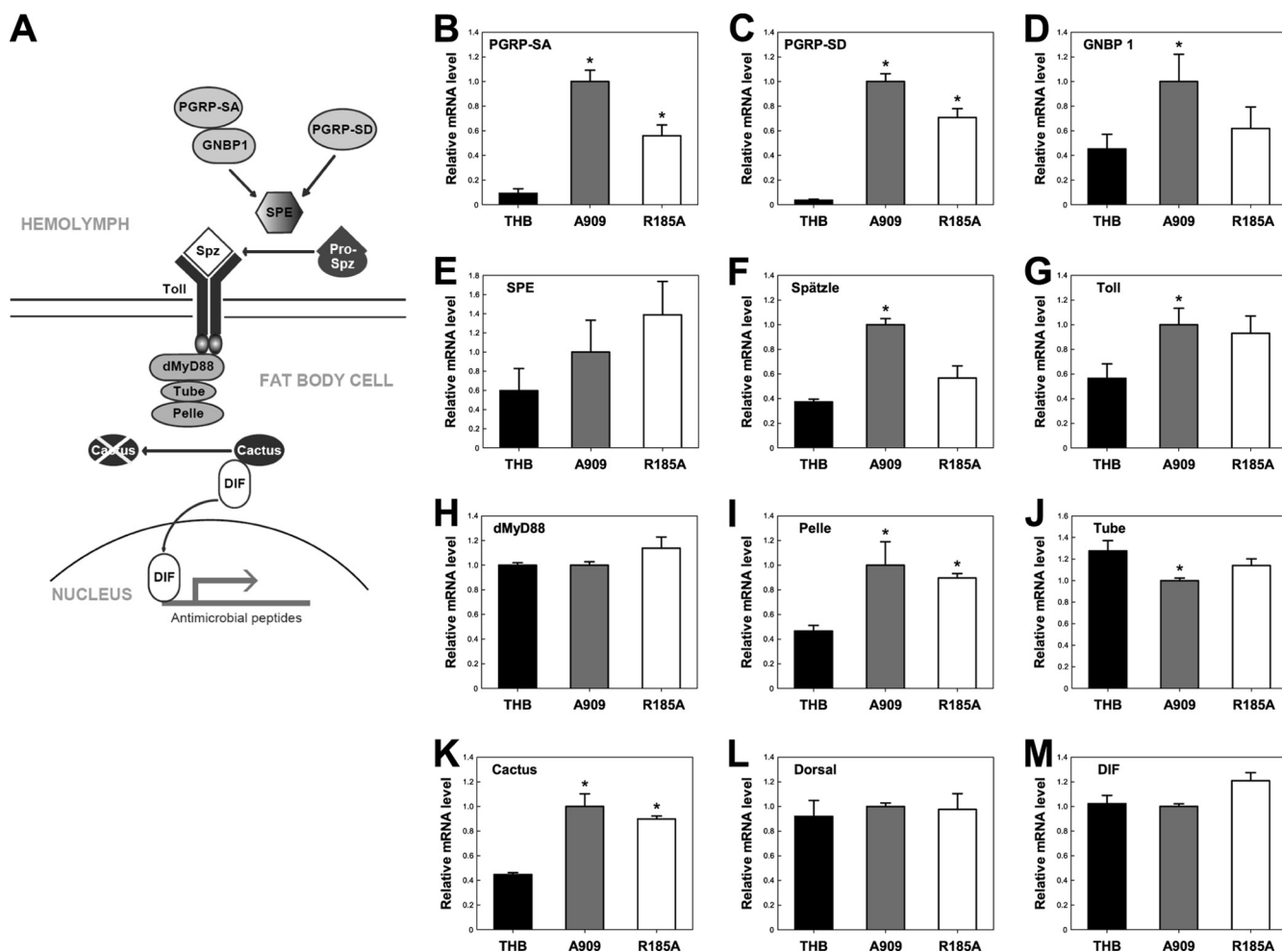


FIG. 2. Gene expression profiling of PGRP-SA, PGRP-SD, and Toll pathway members. (A) Schematic diagram of a subset of the components upstream and downstream of Toll. (B to M) Factors upstream and downstream of Toll that may respond to Gram-positive bacterial infection, PGRP-SA (B), PGRP-SD (C), GNB1 (D), Spätzle-processing enzyme (SPE) (E), Spätzle (F), Toll (G), dMyD88 (H), Pelle (I), Tube (J), Cactus (K), Dorsal (L), and DIF (M), were chosen for quantitative PCR analysis of mRNA levels. The flies were treated by broth injury (black bars), infection with wild-type GBS strain A909 (gray bars), or infection with mutant GBS strain R185A (white bars) and were harvested 24 h postinfection. Two short peptidoglycan recognition protein genes upstream of Toll, PGRP-SA and PGRP-SD, showed dramatic mRNA inducibility in response to GBS infection. *yw* flies were used in the assay whose results are shown here and in the subsequent figures as controls, as they have a genetic background similar to that of transgenic strains. *yw* flies showed AMP induction patterns similar to those of *OregonR* (*OreR*) flies 24 h after infection. Each bar represents the mean and standard error of the mean of relative mRNA levels detected in triplicate independent fly samples. Comparisons of gene transcriptional levels between infected and THB broth-treated control flies were analyzed by Student's *t* test. *, *P* < 0.05.

signaling (8, 34) but are not known to be targets of the Toll pathway. Hereafter, we used *yw* flies instead of *Oregon R* flies as controls because *yw* flies have a genetic background closer to the genetic backgrounds of the transgenic strains used in this study. In *yw* flies, sterile pricking with THB, infection with A909, and infection with R185A were followed after 24 h by AMP transcription patterns comparable to those in *Oregon R* flies (Z. Wang and M. J. Baron, unpublished data). PGRP-SA and PGRP-SD were strongly induced after challenge with A909 (Fig. 2B and C). Moreover, the transcription of several downstream members of the Toll pathway (Gram-negative protein binding 1 [GNBP 1], Spätzle, Toll, Pelle, and Cactus) was also significantly increased after challenge with A909 (Fig. 2D, F, G, I, and K). R185A elicited relatively less transcription of these members. The difference in Spätzle transcription in A909- versus R185A-infected flies

was statistically significant. Other Toll signaling pathway members were constitutively transcribed, with no change in level occurring among the treatment groups (THB versus A909 versus R185A). These pathway members, dMyD88, Tube, Dorsal, and Dorsal-related immunity factor (DIF) (Fig. 2H, J, L, and M), may be regulated at levels other than transcription. Taken together, these data demonstrate that GBS can elicit transcription of the peptidoglycan recognition receptors PGRP-SA and PGRP-SD and of several Toll pathway members.

ACP binding to sulfated host GAGs regulates AMP and PGRP transcription. Complete synthesis and modification of HSPGs requires several genes, i.e., those encoding core proteins, GAG polymerases that initiate and elongate the GAG chains, and sulfation enzymes that modify the chains. To enhance understanding of the effect of the ACP-GAG interaction

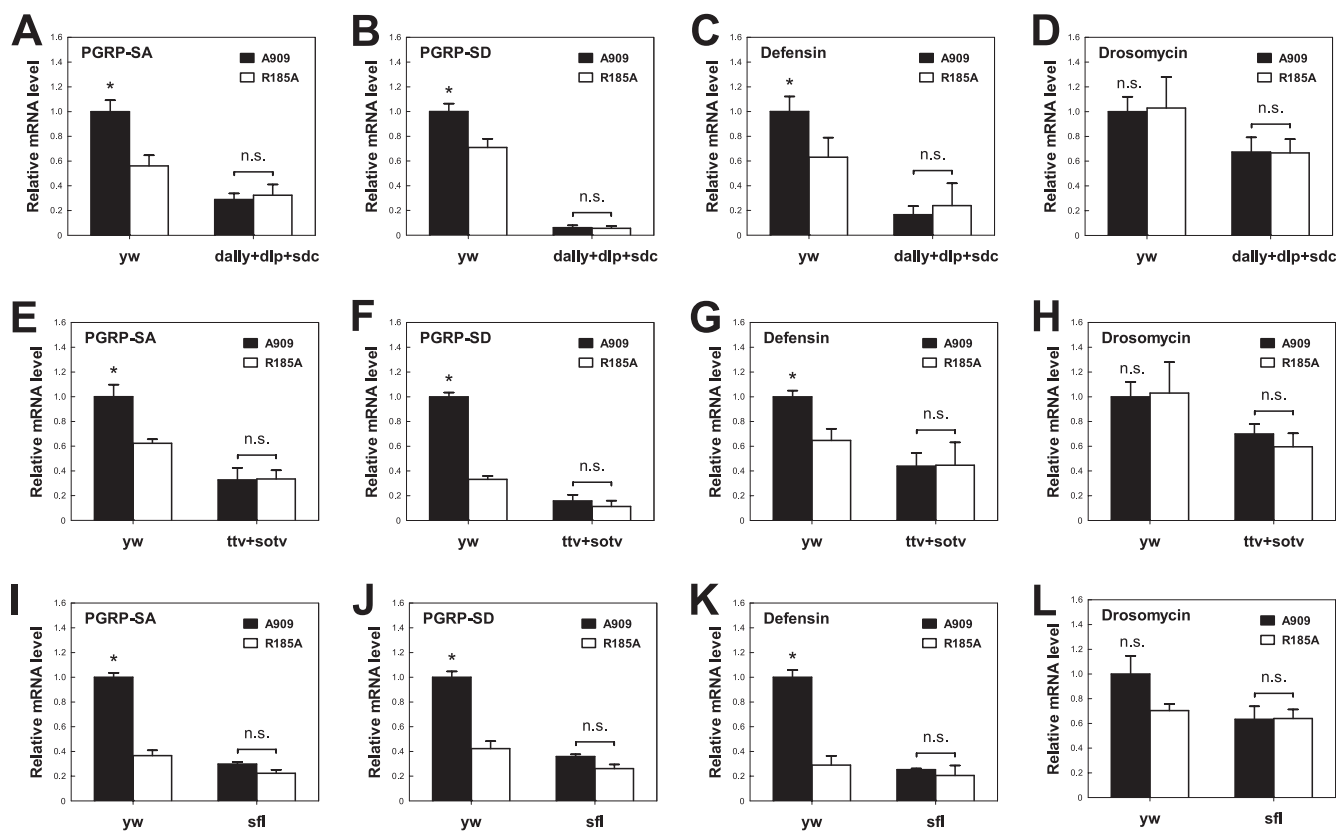


FIG. 3. Induction of PGRPs and AMPs after infection is attenuated in flies with deficient sulfated GAGs/HSPGs. (A to L) The HSPG biosynthesis genes studied here encode core proteins (*dally*, *dlp*, *sdc*; A to D), GAG polymerases (*ttv*, *sotv*; E to H), and the GAG *N*-deacetylase/*N*-sulfotransferase (*sfl*; I to L). All mutants were crossed with *yw* flies to generate heterozygotes because homozygous mutations are lethal. Heterozygous flies were collected 24 h after infection with A909 or R185A. The mRNA levels of PGRP-SA, PGRP-SD, *defensin* (which has exclusive activity against Gram-positive bacteria), and *drosomycin* (which mainly responds to fungal infection) were detected. Each bar represents the mean and standard error of the mean of relative mRNA levels detected in triplicate independent fly samples. Statistical differences between *yw* flies infected with A909 and the other three conditions (*yw* R185A, mutant A909, and mutant R185A) and between infection with A909 versus infection with R185A in mutant flies were compared by Student's *t* test, respectively. *, $P < 0.05$; n.s., no significant difference.

on PGRP and AMP responses, we assessed the transcription of PGRPs and AMPs in fly strains with mutations in the genes encoding HSPG core proteins (*dally*, *dlp*, and *sdc*), GAG polymerases (*ttv* and *sotv*), or the NDST that initiates GAG sulfation (*sfl*). These mutants were generated by crossing mutant flies with *yw* flies, as homozygous mutants in these genes do not survive to adulthood. We found less induction of PGRPs and *defensin* in R185A-infected *yw* flies than in A909-infected *yw* flies (left two bars in each corresponding panel, Fig. 3). In contrast, no significant differences in transcription of PGRPs or *defensin* were detected between A909 and R185A infections in GAG/HSPG mutant flies, *dally*, *dlp*, and *sdc*; *ttv* and *sotv*; and *sfl* (right two bars in each corresponding panel, Fig. 3). Transcription levels of PGRPs and *defensin* were significantly decreased in mutant flies with A909 or R185A infection compared with the transcription levels in *yw* flies with A909 infection. In contrast, the levels of *drosomycin*, an AMP gene that mainly responds to fungal infection, did not differ significantly with the fly genotype or the infecting bacterial strain (Fig. 3D, H, and L). These data indicate that PGRP-SA and PGRP-SD transcription and *defensin* induction require the interaction of ACP and sulfated GAGs on HSPGs.

Interaction of ACP and HSPGs determines the virulence of GBS and promotes bacterial proliferation during fly infection.

Prior studies in *Oregon R* flies demonstrated that ACP-GAG binding is associated with lethal infection and increased GBS burdens after infection (6). To determine the effect of the interaction of ACP and HSPGs on lethal GBS infection in *yw* flies (the background strain used for construction of the transgenic fly lines used in the studies described above and below), we infected *yw* flies and heterozygote offspring of *yw* crossed with GAG/HSPG-deficient flies with A909 and R185A. Heterozygotes were required because homozygous GAG/HSPG mutants do not survive to adulthood. Both wild-type and mutant flies showed greater susceptibility to lethal infection with wild-type GBS strain A909 and less susceptibility to lethal infection with mutant strain R185A. The median survival times after infection with A909 versus R185A were 29.5 h versus 45 h for *yw* flies, 91.5 h versus >91.5 h for *dally dlp sdc* mutant flies, 53 h versus >91.5 h for *ttv sotv* mutant flies, and 48.5 h versus >91.5 h for *sfl* mutant flies (Fig. 4A). Next we assessed total bacterial burdens 24 h after infection. In *yw*, *ttv sotv* mutant, and *sfl* mutant flies, we found lower GBS colony counts after

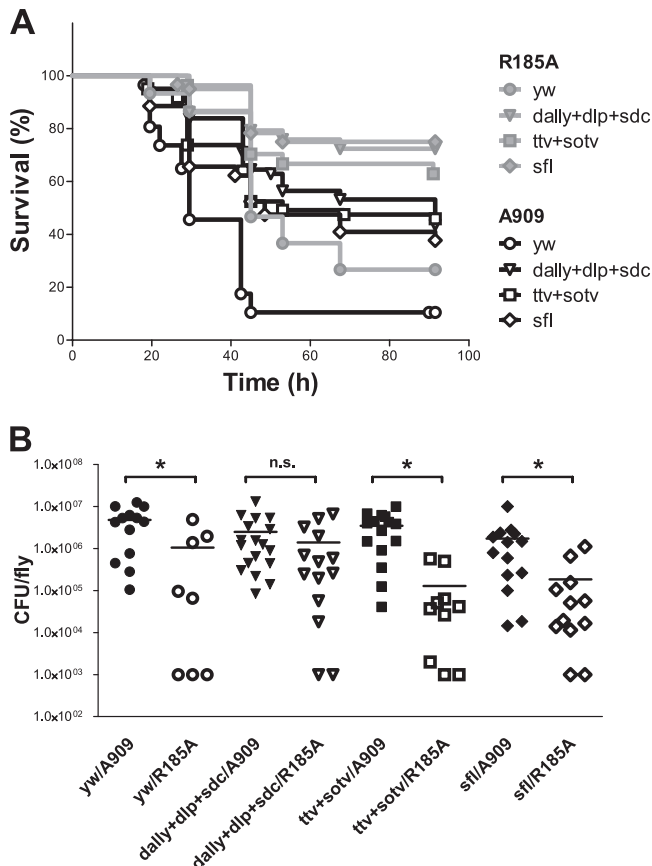


FIG. 4. ACP-GAG binding enhances GBS virulence and increases bacterial burden in wild-type and GAG/HSPG-deficient flies. (A) Both wild-type and GAG/HSPG-deficient flies are more susceptible to lethal infection with A909 than to lethal infection with R185A. The survival curves for yw flies (circles) and *dally dlp sdc* (triangles), *ttv sotv* (squares), and *sfl* (diamonds) mutant flies after infection with A909 (black lines) or R185A (gray lines) are shown. The mutant flies used for the assay whose results are presented here are heterozygous. Plots show a pool of data from at least 30 flies per sample. (B) Infections with R185A are associated with lower bacterial burdens than achieved with A909 infection in all fly strains except those with HSPG core protein mutations. The fly genotype and bacterial infection type are indicated. *, $P < 0.05$; n.s., no significant difference.

infection with R185A than after infection with A909 (Fig. 4B). These data demonstrate that interruption of ACP-GAG binding—via mutation of either bacterial ACP or host sulfated GAGs/HSPGs—attenuated GBS virulence and suppressed bacterial proliferation in *Drosophila*.

Overexpression of Dpp, but not of Hh or Wg, protects flies from GBS infection. Wg, Hh, and the TGF- β homolog Dpp are major secreted signaling molecules controlling cell communication during morphogenesis in *Drosophila* (23). GAGs/HSPGs have been shown biochemically and genetically to interact with each of these morphogens (18). Since GBS can bind to HSPGs and promote lethal infection, we hypothesized that competition between these morphogens and GBS for binding to HSPGs might protect flies from lethal infection. To test this hypothesis, Dpp, Hh, and Wg transgenes were ectopically expressed in the hemocytes by means of a hemocyte-specific *HmlΔ-Gal4* driver. Flies overexpressing Dpp, Hh, or Wg in the

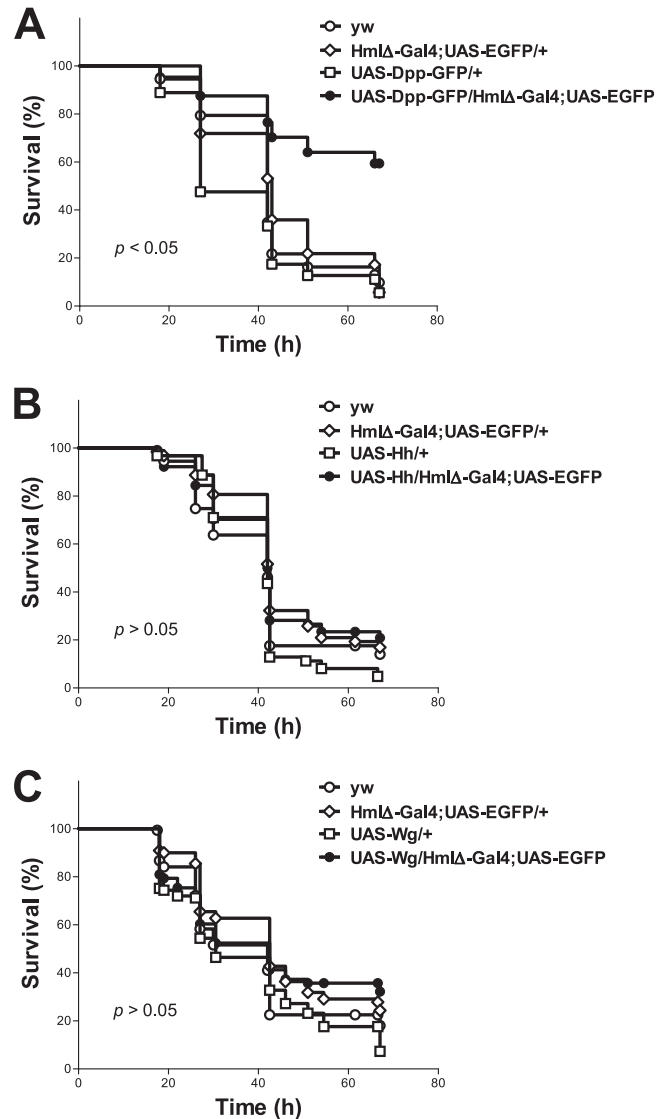


FIG. 5. Overexpression of Dpp but not Hedgehog or Wingless confers resistance to GBS infection. Flies overexpressing Dpp (A), Hh (B), or Wg (C) driven by a hemocyte-specific *HmlΔ-Gal4* driver were infected with A909. Among these three morphogens, only Dpp overexpression (A) significantly protected flies from GBS infection. Plots show a pool of data from at least 30 flies per treatment. The *Gal4* and *UAS* lines were crossed with yw flies as controls. The survival curves of yw, *Gal4* controls, *UAS* controls, and overexpression groups are indicated as open circles, open diamonds, open squares, and filled circles, respectively. A log rank test was used for each comparison (P values are shown in the plot).

hemocytes developed normally, with no obvious defects after they were hatched, and survived sterile prick injury with THB. After infection with wild-type GBS strain A909, flies overexpressing Dpp in the hemocytes had higher survival rates over time than did control flies (Fig. 5A). In contrast, overexpression of Hh or Wg had no such protective effect (Fig. 5B and C). Thus, overexpression of Dpp, but not of Hh or Wg, rescued flies from lethal GBS infection, a result compatible with the hypothesis that overexpressed Dpp may interfere with the ability of GBS to interact with host HSPGs.

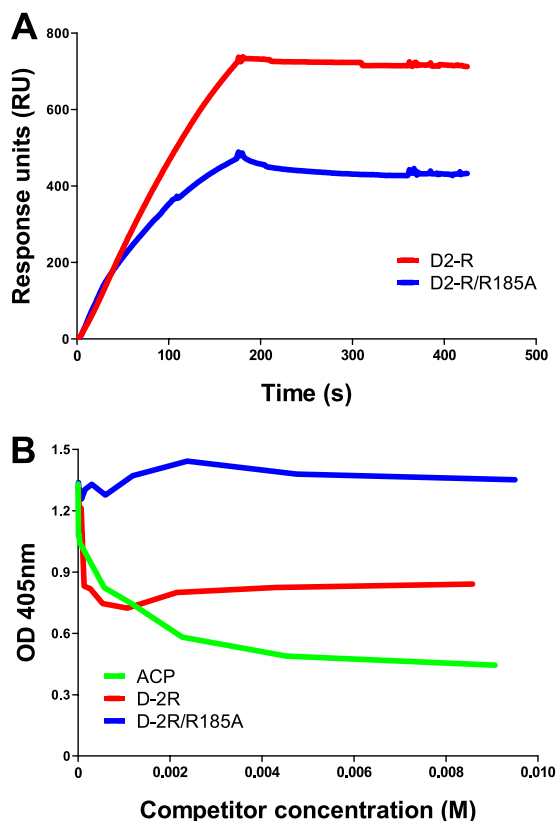


FIG. 6. ACP competes with Dpp to bind to heparin, a prototype sulfated GAG. (A) Binding affinity of ACP constructs to immobilized HS. Interaction was monitored by plotting the increase in response over time. Wild-type D2-R bound to HS with stronger affinity than did D2-R/R185A. (B) Full-length ACP and D2-R proteins specifically compete with Dpp for binding to heparin. A competition binding assay was performed, in which wells were coated with heparin and then incubated with a fixed concentration ($5 \times 10^{-3} \mu\text{g}/\mu\text{l}$) of *Drosophila* Dpp protein and various concentrations of competitor proteins. Competitor proteins include ACP, D2-R, and D2-R/R185A with reduced heparin-binding affinity. Bound Dpp was detected with primary/secondary antibodies and alkaline phosphatase substrate. OD, optical density.

ACP competes with Dpp to bind to heparin *in vitro*. The improved survival of Dpp-overexpressing flies prompted us to evaluate whether ACP might compete directly with Dpp for binding to GAGs/HSPGs. We tested this hypothesis through a competition ELISA in which plate wells were coated with the prototype GAG heparin and then incubated with a fixed (saturating) concentration of recombinant Dpp ($5 \times 10^{-3} \mu\text{g}/\mu\text{l}$) and with various amounts of recombinant ACP or D2-R (a truncated protein construct including the GAG-binding region of ACP) (3). We used Dpp-specific antibody to measure the amount of bound Dpp in the presence of competing GAG-binding proteins. As a control for these studies, we used D2-R/R185A, a mutant version of ACP in which a charge-neutralizing mutation in the GAG-binding region is associated with reduced cell-binding activity, reduced GAG-binding ability in dot blot assays (5), and reduced GAG-binding affinity for immobilized HS, as measured by SPR (Fig. 6A). As the concentration of ACP or D2-R competitor increased, the amount of heparin-bound Dpp declined before it reached a plateau at an

ACP concentration of $2.27 \times 10^{-3} \text{ M}$ and a D2-R concentration of $1.07 \times 10^{-3} \text{ M}$. In contrast, similar concentrations of D2-R/R185A had little effect on the levels of bound Dpp (Fig. 6B). These data demonstrate that ACP and D2-R with intact GAG-binding ability can compete with Dpp for binding to heparin *in vitro*.

The GAG-binding region of Dpp is essential for protection from GBS infection but not for suppression of PGRP-SA or -SD. A 7-amino-acid sequence at the N terminus of Dpp interacts with heparin and with HSPGs in *Drosophila* (1). We hypothesized that this GAG-binding domain of Dpp is essential for the effects of Dpp overexpression in enhancing host survival after GBS infection. To test this hypothesis, we overexpressed wild-type Dpp or a truncated Dpp lacking the GAG-binding domain (Dpp^{ΔN}, indicated as ΔNdpp) in the hemocytes, using the HmlΔ-Gal4 driver. Consistent with the results shown in Fig. 5A, flies overexpressing wild-type Dpp survived significantly longer after A909 infection than did any of the control flies (Fig. 7A). In contrast, the survival of flies overexpressing the truncated form of Dpp was similar to that of the control flies after A909 infection (Fig. 7A). The protection provided by overexpression of wild-type Dpp was lost when infection was performed with GBS strain R185A, which has reduced GAG-binding ability (Fig. 7B). These data suggest that Dpp-GAG-binding activity is required for Dpp-mediated protection of flies from lethal GBS infection.

We measured mRNA levels of PGRP-SA and PGRP-SD as well as AMPs in these Dpp-overexpressing flies 24 h after GBS infection. Both Dpp- and ΔNdpp-overexpressing flies showed similarly decreased PGRP-SA and PGRP-SD induction compared with that for the control flies (Fig. 7C and D). Although mRNA levels of *defensin* and *dipterocin* were lower in both Dpp- and ΔNdpp-overexpressing flies than in control flies, Dpp-overexpressing flies exhibited relatively lower AMP expression than did ΔNdpp-overexpressing flies (Fig. 7E and F). These data indicate that the binding of Dpp to GAGs/HSPGs plays a role in protection from GBS infection and in AMP transcription but not in suppression of PGRP-SA and PGRP-SD transcription.

DISCUSSION

Sulfated GAGs on the surface of mammalian cells play multiple important roles, including serving as attachment sites for numerous pathogens. The mechanisms by which host GAGs may influence the outcome of host-pathogen interactions are incompletely understood. To address this issue, we took advantage of the relatively simple genome and well-developed approaches for genetic manipulation in *Drosophila*. Importantly, the GAG/HSPG synthesis machinery, GAG repertoire, and innate immune response pathways in *Drosophila* are strongly homologous to those in mammals. We found that (i) AMP transcription correlated with ACP-GAG binding, (ii) overexpression of wild-type but not of GAG-binding-deficient Dpp in the hemocytes enhanced host survival after infection, (iii) ACP competed with Dpp for GAG binding, and (iv) both wild-type and GAG-binding-deficient Dpp were able to modulate PGRP-SA and PGRP-SD transcription, while Dpp-GAG binding appeared to alter AMP transcription through pathways independent of PGRP-SA and PGRP-SD signaling. These re-

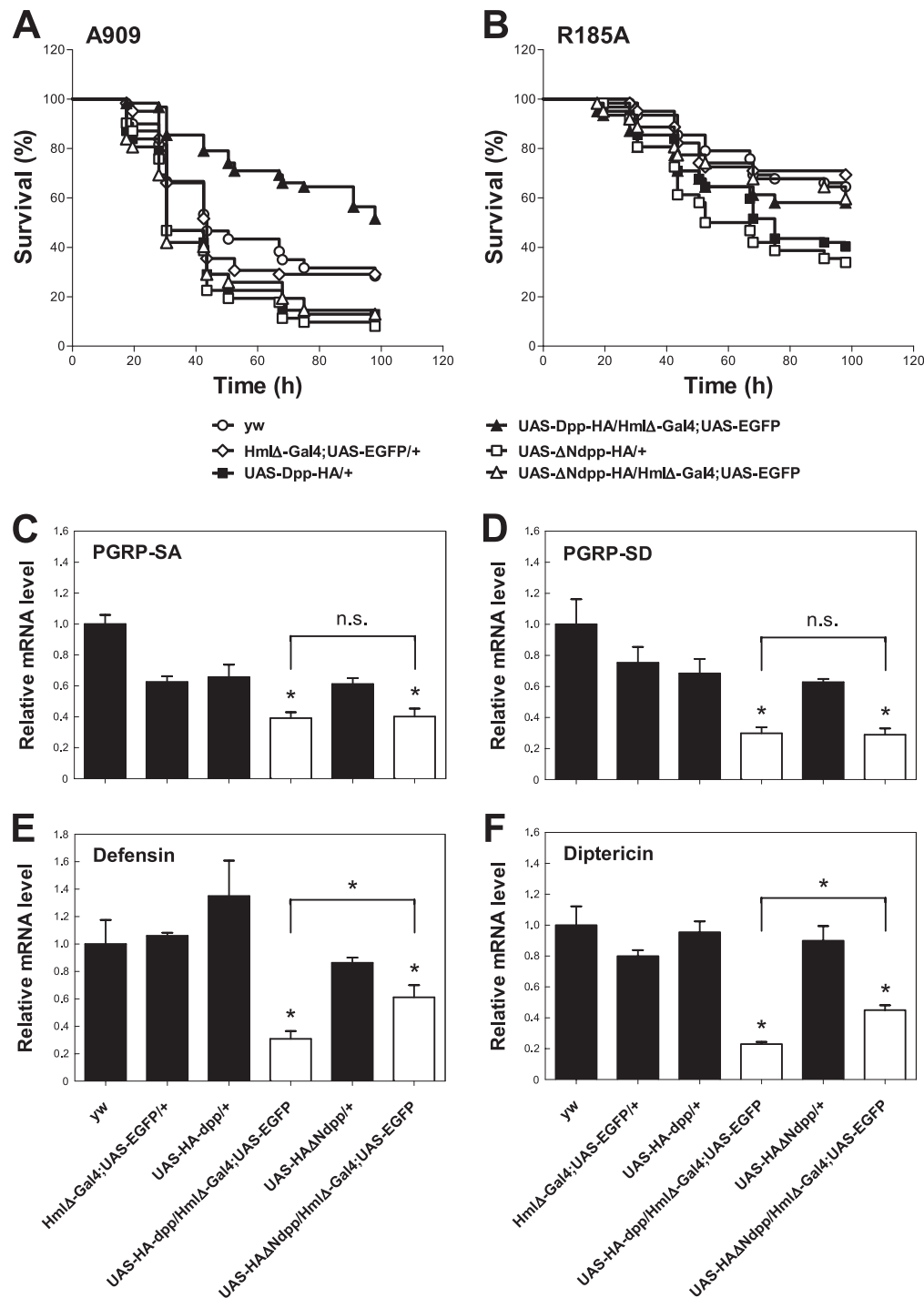


FIG. 7. Effects of the Dpp heparin-binding region in fly survival and immunosuppression. (A and B) Dpp^{ΔN} (indicated ΔNdp) is a truncated form of Dpp lacking seven N-terminal amino acid residues that are essential for interacting with heparin and HSPGs (1). Overexpression of the truncated Dpp and its counterpart wild-type Dpp was driven by HmlΔ-Gal4. Overexpression of wild-type Dpp protected flies from lethal A909 infection ($P < 0.01$), while overexpression of Dpp^{ΔN} did not ($P > 0.05$) (A). In the infection with GAG-binding mutant GBS strain R185A, overexpression of neither wild-type Dpp nor Dpp^{ΔN} showed protection compared with results for the control groups ($P > 0.05$) (B). The survival assays with Dpp- and ΔNdp-overexpressing flies were performed and the results were analyzed together in duplicate experiments. Plots show a pool of data from at least 30 flies per sample. The Gal4 and UAS lines were crossed with *yw* flies as controls. The survival curves of *yw*, Gal4 controls, UAS controls, and overexpression groups are indicated as circles, diamonds, squares, and triangles, respectively. Detailed genotypes are provided below panels A and B. A log rank test was used for each comparison. (C to F) Transcription of PGRP-SA (C), PGRP-SD (D), *defensin* (E), and *dipthericin* (F) was measured in these flies 24 h after infection. Results for flies overexpressing Dpp are shown as white bars, and those for control flies are shown as black bars. The detailed genotypes in panels C to F are provided below panels E and F. Each bar represents the mean and standard error of the mean of relative mRNA levels detected in triplicate independent fly samples. Statistical differences between overexpression flies versus control flies and between overexpression of wild-type versus truncated Dpp were compared by Student's *t* test, respectively. *, $P < 0.05$; n.s., no significant difference.

sults have several implications for the role of GAGs in host responses to pathogens.

First, Dpp, a GAG-binding *Drosophila* homolog of TGF- β , modulates transcription of host AMPs, which constitute a major arm of the innate immune defense system in *Drosophila* and which also have mammalian homologs. Previous microarray studies demonstrating that Dpp expression gradually increases by nearly 2-fold during the course of bacterial infection further support a role for Dpp in host responses to infection (14, 15).

Our data demonstrating that Dpp can suppress AMP transcription and enhance host survival after infection suggest parallel functions of TGF- β -like structures in *Drosophila* and in mammals; mammalian TGF- β is known to have immune response-regulating effects (9, 24, 37) that may limit excessive immune activation detrimental to the host. There is some evidence that chronic—but not acute—AMP activation is detrimental to survival of the *Drosophila* host over time (28). These data raise the possibility that by suppressing AMP transcription, Dpp may limit excessive immune activation that is detrimental to *Drosophila* host survival. Importantly, the positively charged amino acids at the N terminus of Dpp that confer its GAG-binding activity are highly conserved in the mammalian Dpp homolog, bone morphogenetic protein 4 (BMP-4) (1).

As acute increases in AMP transcription seem not to be detrimental to *Drosophila* hosts, Dpp overexpression may modify immune system development and enhance host survival through mechanisms other than modulation of AMP transcription. For example, Dpp has been reported to play a role in the regulation of blood cell homeostasis and defenses against oral infection of *Drosophila* with *Salmonella enterica* serovar Typhimurium (17). In these studies, the overall number of blood cells was higher in *dpp* mutant flies than in wild-type controls, but lamellocyte numbers were deficient in the mutant flies. These data suggest that Dpp overexpression should lead to lower numbers of blood cells and less effective immune responses, perhaps with an excess of lamellocytes. In contrast, we found that Dpp overexpression protected flies from infection. To assess the effects of Dpp overexpression on blood cell numbers, we used a green fluorescent protein (GFP) construct combined with Hml Δ -Gal4 to visualize GFP-labeled cells, which represent ~95% of adult blood cells (12). We found no significant difference in GFP signals among Gal4 control, Dpp-overexpressing, and truncated Dpp-overexpressing flies. This result indicated that overexpression of Dpp has no major effect on numbers of adult hemocytes (Wang and Baron, unpublished). Whether Dpp overexpression affects other aspects of immunity is an important question for future studies.

Second, we found that ACP, a bacterial GAG-binding protein, could compete with Dpp for binding to GAGs and that a GBS strain expressing ACP induced AMP transcription during infection. In contrast, an ACP variant with reduced GAG-binding affinity due to the R185A mutation could not compete with Dpp for binding to GAGs, while a GBS strain expressing this mutant protein induced less AMP transcription. In addition, hosts with reduced levels of sulfated GAGs had lower AMP induction levels than did wild-type hosts. The correlation of host AMP responses with affinity of ACP-GAG binding further supports the hypothesis that GAG binding significantly affects AMP transcription.

In planning these studies, we had hypothesized that acute AMP inductions would defend the host against infection. Thus, the finding that AMP induction levels were higher in infections associated with worse host outcomes was somewhat unexpected but, in fact, correlated with the findings of other studies (2, 12, 19). As the literature suggests that acute AMP inductions are not directly harmful to the host (28), other possible explanations must be considered.

One possible explanation for the association of higher-level AMP inductions with worse infection outcomes is that AMP inductions occur in proportion to bacterial load, such that higher CFU loads induce higher levels of AMP transcription. However, some data suggest that there is no simple correlation between CFU load and AMP transcription level. For example, wild-type and HSPG-deficient flies infected with A909 exhibited significant differences in AMP transcription levels, despite similar CFU loads. Further, the AMP transcription level was similar in HSPG-deficient flies infected with either A909 or R185A, despite significant differences in CFU loads. Finally, we found similar AMP transcription levels in flies infected with bacterial inocula that were 5-fold higher than the doses used in our standard infection experiments; that is, a 5-fold increase in A909 dose was associated with AMP transcription levels similar to those after the standard A909 dose (see Fig. S1 in the supplemental material). Therefore, differences in CFU load appear not to account for differences in AMP induction levels.

Rather, the association of higher levels of AMP induction with worse host outcomes suggests that AMP gene expression levels may reflect more efficient AMP induction by GBS organisms that use ACP-GAG binding to enter host cells most effectively. In this case, AMP transcription levels may be unrelated to the enhanced host survival that we observed with either the mutant GBS strain or Dpp-overexpressing flies due to one of the following possibilities: (i) AMP inductions may occur too late in the GBS infection course to protect the host. Studies in beetles have suggested that rather than acting as primary host defense effectors, induced antimicrobial compounds may primarily protect the insect against the bacteria that survive primary defenses, such as phagocytosis by hemocytes (19). Supporting the possibility that earlier AMP inductions might enhance host survival after GBS infection, the infection of flies constitutively overexpressing defensin in hemocytes or fat body cells demonstrated that expression of this AMP constitutively and at high levels protected flies against a fatal outcome (see Fig. S2 in the supplemental material). (ii) AMP responses are not effective at protecting hosts from lethal infection, perhaps because secreted AMPs cannot access GBS organisms that have entered host cells (5, 10).

The effects of AMP transcription timing, ability of AMPs to access intracellular GBS, and potential role of autophagy or other immune mechanisms in defense against GBS merit future study.

While raising these questions, our data expand our understanding of GAG binding in the pathogenesis of and host response to GBS and other microbial infections. Specifically, the data suggest that during GBS infection, ACP present on the bacterial surface not only promotes bacterial adhesion to host cell surface HSPGs (4) and mediates bacterial entry into host cells (5, 10) but also leads to induction of host AMP transcription, perhaps by disrupting the binding of Dpp to

sulfated GAGs/HSPGs. Conversely, host upregulation of either existing extracellular Dpp or newly produced Dpp may promote host survival by interfering with ACP-GAG/HSPG binding.

The mechanism by which Dpp suppresses AMP transcription requires further study. One possibility is that the Dpp pathway regulates the inducibility of PGRP-SA and PGRP-SD, as a significant decrease in PGRP-SA and PGRP-SD transcription was observed in Dpp-overexpressing flies. Such regulation may occur via GAG-independent mechanisms, since it occurred in flies overexpressing either wild-type or mutant Dpp. The effects may occur via the Imd pathway, as transcription levels of PGRP-SA, PGRP-SD, and all tested AMPs were reduced in Imd mutant flies compared to those in wild-type flies after infection with A909 or R185A (see Fig. S3 in the supplemental material). These data correlate with those in other reports suggesting that the Imd pathway may contribute to regulation of PGRP-SA and PGRP-SD (15). Since HSPGs are among several recognized receptors for Dpp (18), future studies of the potential role of other Dpp receptors on AMP regulation are needed. The determinants and impact of Dpp binding to its other receptors, such as Tkv and Punt, are not yet clear. However, these interactions are undoubtedly more complex in the context of infection with pathogens expressing GAG-binding virulence structures such as ACP.

In summary, the events that occur during infection result from the interaction of two complex systems—pathogen and host—each of which has developed strategies to survive the other's responses. Together with previous studies, our results support the following model for the molecular events that take place during the host-GBS interaction. Bacteria adhere to host cell surface GAGs/HSPGs and enter host cells via the binding of the virulence factor ACP to these GAGs/HSPGs. This binding also modulates activation of AMP production, which may be triggered by PGRP recognition of GBS. Meanwhile, Dpp suppresses AMP transcription, competes with ACP for HSPG binding, and has the potential to protect the host from lethal GBS infection. The contribution of AMP transcription to host survival is not yet clear. Further studies must determine why higher-level AMP induction does not correspond with improved host survival. Specifically, it will be important to explore the possibilities that AMP inductions fail to protect hosts from infection because (i) AMP transcription may occur too late in the process of infection or (ii) secreted AMPs are unable to kill wild-type GBS because these organisms enter cells. Another area for further exploration is whether the Dpp pathway suppresses responses (including AMP transcription) to prevent hyperactivation of innate immune effectors.

These data suggest that host cell surface matrix HSPGs play a crucial role in GBS virulence and host defense in *Drosophila*. An understanding of signaling pathways regulated by HSPGs in the context of host responses to GAG-binding pathogens in *Drosophila* and in mammals may lead to the identification of new targets and approaches for preventing infection and/or enhancing host defenses and survival.

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